PRACTICAL 13
SSH Configuration

Aim
To configure an SSH server and access a remote RHEL system using password and key-based authentication.

A. Install SSH server
sudo dnf install openssh-server
Start:
sudo systemctl enable --now sshd
Check:
systemctl status sshd

B. Find IP
ip addr

C. SSH Login
From another Linux machine:
ssh username@IP_ADDRESS
Example:
ssh labuser@192.168.56.20

D. Generate SSH Key
On the client:
ssh-keygen -t ed25519
When prompted, enter a **passphrase**.

E. Copy Public Key
ssh-copy-id labuser@192.168.56.20
Login:
ssh labuser@192.168.56.20
The private key should remain protected and should never be copied to the server.

PRACTICAL 14
Apache Web Server

Aim
To configure a basic Apache web server and create a virtual host.

Install Apache
sudo dnf install httpd
Start:
sudo systemctl enable --now httpd
Check:
systemctl status httpd

Create Web Page
sudo vi /var/www/html/index.html
Enter:

```
<!DOCTYPE html>
<html>
<head>
   <title>TYIT Linux Practical</title>
</head>
<body>
   <h1>Welcome to TYIT RHEL Web Server</h1>
   <p>Apache Web Server is working successfully.</p>
</body>
</html>
```

Test locally:
curl http://localhost

Firewall
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
Check:
sudo firewall-cmd --list-services

Apache Configuration Files
Important locations:
/etc/httpd/
/etc/httpd/conf/httpd.conf
/etc/httpd/conf.d/
/var/www/html/
/var/log/httpd/
Check:
httpd -t
Expected:
Syntax OK

Virtual Host

Create:

sudo vi /etc/httpd/conf.d/tyit.conf

Example:

```
<VirtualHost *:80>
    ServerName tyit.local
    DocumentRoot /var/www/tyit

    <Directory /var/www/tyit>
        AllowOverride None
        Require all granted
    </Directory>
</VirtualHost>
```

Create directory:

sudo mkdir -p /var/www/tyit

Create page:

echo "TYIT Virtual Host" | sudo tee /var/www/tyit/index.html

Check configuration:

sudo httpd -t

Restart:

sudo systemctl restart httpd

PRACTICAL 15
Firewall Configuration

Aim
To configure Linux firewall using firewalld and study the relationship with iptables/nftables.

A. Check firewalld
sudo systemctl status firewalld

B. Start firewall
sudo systemctl enable --now firewalld

C. Check active zone
sudo firewall-cmd --get-active-zones

D. List services
sudo firewall-cmd --list-services

E. Allow HTTP
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
Verify:
sudo firewall-cmd --list-services

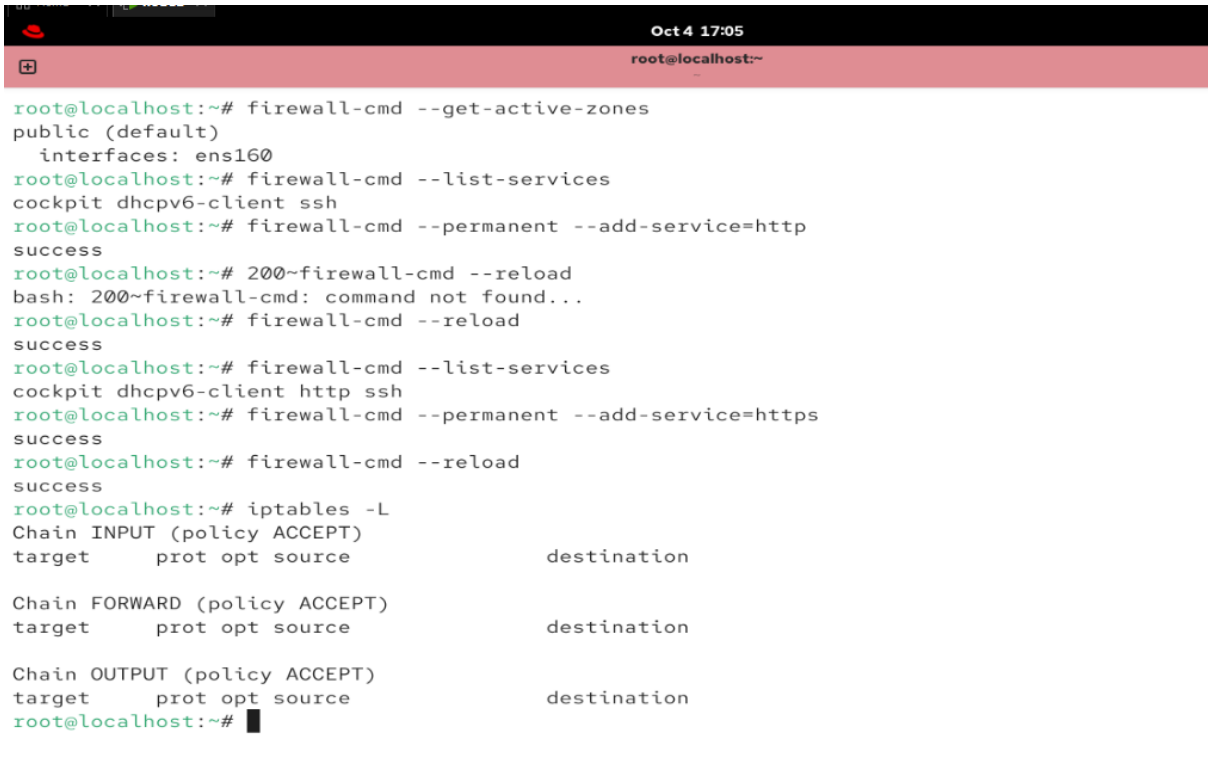
F. Allow HTTPS
sudo firewall-cmd --permanent --add-service=https
sudo firewall-cmd --reload

G. Remove service
sudo firewall-cmd --permanent --remove-service=http
sudo firewall-cmd --reload
RHEL documentation recommends predefined firewalld services for allowing required service traffic.

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iptables
Check availability:
sudo iptables -L
On modern RHEL, students should understand that the underlying packet-filtering architecture is based around nftables, while firewalld provides the normal high-level management interface.


root@localhost:~
tmpfs 168M 120K 168M 1% /run/user/0
/dev/sr0 11G 11G 0 100% /run/media/root/RHEL-10-2-BaseOS-x86_64
root@localhost:~#
root@localhost:~# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
Active: active (running) since Sun 2026-10-04 16:33:45 IST; 27min ago
Invocation: ec2a832071674891a7204d9812e42f17
Docs: man:firewalld(1)
Main PID: 1212 (firewalld)
Tasks: 2 (limit: 10316)
Memory: 43.6M (peak: 48.5M)
CPU: 841ms
CGroup: /system.slice/firewalld.service
└─1212 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid
Oct 04 16:33:43 localhost systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon
Oct 04 16:33:45 localhost systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
root@localhost:~# systemctl enable --now firewalld
root@localhost:~# firewall-cmd --get-active-zones
public (default)
interfaces: ens160
root@localhost:~# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@localhost:~# firewall-cmd --permanent --add-service=http
success


Oct 4 17:05
root@localhost:~
root@localhost:~# firewall-cmd --get-active-zones
public (default)
interfaces: ens160
root@localhost:~# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@localhost:~# firewall-cmd --permanent --add-service=http
success
root@localhost:~# 200~firewall-cmd --reload
bash: 200~firewall-cmd: command not found...
root@localhost:~# firewall-cmd --reload
success
root@localhost:~# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
root@localhost:~# firewall-cmd --permanent --add-service=https
success
root@localhost:~# firewall-cmd --reload
success
root@localhost:~# iptables -L
Chain INPUT (policy ACCEPT)
target prot opt source destination
Chain FORWARD (policy ACCEPT)
target prot opt source destination
Chain OUTPUT (policy ACCEPT)
target prot opt source destination
root@localhost:~#